EEG-Based Emotion Recognition

using Deep Convolutional Neural Network

Dataset: GAMEEMO

Classification Accuracy: 97.06%

Emotion Classes: Boring | Calm | Horror | Happy

Domain: Affective Computing & Brain-Computer Interfaces

Technology: Python | PyTorch | Deep CNN

Year: 2025

Abstract

Emotion recognition using Electroencephalography (EEG) signals is a key area in affective computing and brain-computer interface systems. This project presents a Deep Convolutional Neural Network (CNN) framework for classifying human emotions using EEG signals from the GAMEEMO dataset. The system processes 14-channel EEG signals and classifies them into four emotional states: Boring, Calm, Horror, and Happy.

A preprocessing pipeline involving band-pass filtering, normalization, and segmentation is applied to enhance signal quality. The proposed Deep CNN model extracts hierarchical spatiotemporal features, achieving a high classification accuracy of 97.06%. The model outperforms traditional approaches such as LSTM and baseline CNN, demonstrating its effectiveness for real-time emotion recognition applications in healthcare, gaming, and human-computer interaction.

1. Introduction

Emotions significantly influence human cognition, communication, and decision-making. Automatic emotion recognition can improve applications such as mental health monitoring, adaptive learning systems, and intelligent interfaces. As the demand for intelligent human-computer interaction grows, the need for reliable and real-time emotion detection systems becomes increasingly critical.

Among various physiological signals, EEG is widely used due to its direct measurement of brain activity, high temporal resolution, and relatively low cost. However, EEG signals are complex, noisy, and vary significantly across individuals, making emotion recognition a challenging task.

Recent advancements in deep learning have substantially improved EEG-based emotion recognition. This project focuses on using a Deep CNN model to efficiently extract spatiotemporal features from EEG data and improve emotion classification accuracy compared to traditional machine learning and recurrent approaches.

2. Problem Statement

EEG-based emotion recognition presents several unique challenges that limit the effectiveness of conventional approaches:

- High noise levels and physiological artifacts in EEG signals

- Non-linear and non-stationary nature of brain activity signals
- Significant inter-subject variability in EEG patterns
- Limited classification performance of traditional ML models
- Difficulty in capturing multi-scale spatiotemporal features simultaneously

This project aims to design a robust deep learning model that can accurately classify emotions from EEG signals and systematically overcome these challenges by leveraging hierarchical feature extraction through a deep convolutional architecture.

3. Literature Survey

Several deep learning and machine learning approaches have been proposed for EEG-based emotion recognition. A comparative summary of key prior works is presented below:

Model	Key Contribution	Accuracy	Limitation
LSTM	Captures temporal dependencies in sequential EEG signals	~91%	Moderate accuracy
CNN	Good spatial feature extraction from multi-channel EEG	~89%	Limited temporal understanding
Bi-LSTM	Bidirectional sequence learning for better context	~87%	High complexity
KNN / SVM	Traditional ML with handcrafted EEG features	<85%	Poor generalization

The limitations of prior work include poor generalization across subjects and inability to capture multi-scale features effectively. The proposed Deep CNN model addresses these gaps through hierarchical feature extraction and a carefully designed training pipeline.

4. Scope and Objectives

4.1 Scope

- EEG-based emotion recognition from physiological signals
- Brain-computer interface (BCI) system design
- Healthcare applications including mental health monitoring
- Adaptive human-computer interaction systems

4.2 Objectives

1. Effectively preprocess raw EEG signals to remove noise and artifacts
2. Design and implement a Deep CNN model optimized for EEG data
3. Classify emotions into four distinct categories: Boring, Calm, Horror, and Happy
4. Achieve higher classification accuracy than existing baseline models
5. Evaluate model performance using standard metrics: Accuracy, Precision, Recall, and F1-Score

5. System Architecture

The overall system architecture consists of a sequential pipeline from raw EEG data acquisition to classified emotion output. The pipeline integrates preprocessing, deep feature extraction, and classification stages as illustrated below:

EEG Data Acquisition → Signal Preprocessing → Deep CNN Feature Extraction → Emotion Classification

Each component of the pipeline plays a critical role: preprocessing ensures signal quality, the CNN layers extract hierarchical spatiotemporal features, and the final dense layer with softmax activation produces a probability distribution over the four emotion classes.

6. Methodology

6.1 Dataset: GAMEEMO

Dataset: GAMEEMO (Game-Induced Emotion EEG Dataset)
Channels: 14-channel EEG signals
Emotion Classes: Boring, Calm, Horror, Happy (4 classes)
Stimuli: Gaming scenarios designed to elicit distinct emotional responses

6.2 Preprocessing Pipeline

The following preprocessing steps are applied to ensure signal quality before model training:

- Band-pass Filtering (0.5 Hz - 45 Hz): Removes low-frequency drift, high-frequency noise, and unwanted artifacts including muscle activity

- Z-score Normalization: Standardizes EEG signal amplitudes across channels to zero mean and unit variance, reducing inter-subject variability
- Sliding Window Segmentation (50% overlap): Divides continuous signals into fixed-length segments, improving temporal learning and effectively augmenting the dataset size

Training Set	Validation Set	Test Set
68%	12%	20%

6.3 Deep CNN Model Architecture

The proposed Deep CNN model consists of four progressive convolutional blocks followed by global average pooling and fully connected layers:

Block	Layer Type	Filters	Kernel	Additional
Input	EEG Signal (14 channels)	-	-	Preprocessed input
Block 1	Conv1D + BatchNorm + ReLU	64	7	MaxPool, Dropout 0.25
Block 2	Conv1D + BatchNorm + ReLU	128	5	MaxPool, Dropout 0.25
Block 3	Conv1D + BatchNorm + ReLU	256	3	MaxPool, Dropout 0.3
Block 4	Conv1D + BatchNorm + ReLU	512	3	MaxPool, Dropout 0.4
GAP	Global Average Pooling	-	-	Dimension reduction
FC	Dense + Softmax	-	-	4-class output

Each convolutional block progressively increases filter depth to capture more complex and abstract EEG feature representations. Batch Normalization stabilizes training, Dropout prevents overfitting, and Max Pooling reduces spatial dimensionality.

6.4 Training Configuration

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001

Hyperparameter	Value
Loss Function	Categorical Cross-Entropy
Batch Size	32
Activation (Hidden)	ReLU
Activation (Output)	Softmax
Hardware	GPU Accelerated

7. Evaluation Metrics

Model performance is evaluated using the following standard classification metrics:

- Accuracy = (TP + TN) / Total — Overall fraction of correctly classified samples
- Precision = TP / (TP + FP) — Proportion of positive predictions that are correct
- Recall = TP / (TP + FN) — Proportion of actual positives correctly identified
- F1-Score = 2 x (Precision x Recall) / (Precision + Recall) — Harmonic mean balancing precision and recall

These metrics provide a comprehensive evaluation of classification performance, particularly in multi-class scenarios where class imbalance may exist. All four metrics are reported per-class and as weighted averages.

8. Results and Discussion

8.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Deep CNN (Proposed)	**97.06%**	**0.97**	**0.97**	**0.97**
LSTM	91.18%	0.91	0.91	0.91
CNN (Baseline)	89.92%	0.90	0.89	0.89
Bi-LSTM	87.39%	0.87	0.87	0.87

8.2 Key Findings

- The proposed Deep CNN model achieves the highest accuracy of 97.06%, outperforming all compared models

- Subject-wise accuracy reaches 99.24%, demonstrating strong generalization across individual participants
- Precision, Recall, and F1-Score are all approximately 0.97, indicating balanced performance across emotion classes
- The hierarchical convolutional design effectively captures both fine-grained and high-level EEG features
- Minimal classification errors observed in the confusion matrix, especially for Calm and Happy classes

The results confirm that increasing model depth with progressive filter sizes (64 to 512) significantly improves EEG feature representation. The use of Batch Normalization and strategic Dropout rates (0.25 to 0.4) contributes to both training stability and generalization performance.

9. Advantages and Limitations

9.1 Advantages

- High classification accuracy of 97.06% with robust cross-subject performance
- Effective hierarchical spatiotemporal feature extraction via deep convolutional blocks
- Faster convergence and training efficiency with GPU acceleration
- Suitable for real-time emotion recognition applications
- Generalizes well across subjects with minimal fine-tuning required

9.2 Limitations

- Requires significant computational resources due to model depth and complexity
- Performance is sensitive to the quality and consistency of input EEG data
- EEG signals remain inherently noisy and subject-dependent
- Current evaluation is limited to the GAMEEMO dataset; broader generalization is untested

10. Tools and Technologies

Category	Details
Programming Language	Python 3.x
Deep Learning Framework	PyTorch
Numerical Computing	NumPy, Pandas
Signal Processing	SciPy (band-pass filtering)

Category	Details
Visualization	Matplotlib, Plotly
Hardware Acceleration	GPU (CUDA-enabled)
Model Architecture	Deep Convolutional Neural Network (CNN)

11. Future Work

Several enhancements are planned to further improve the system's capabilities and applicability:

6. Real-time emotion detection integration with streaming EEG hardware
7. Deployment with portable, wearable EEG devices for practical field use
8. Exploration of Transformer-based architectures (e.g., EEG-Conformer, EEGFormer)
9. Cross-dataset validation to improve and measure generalization beyond GAMEEMO
10. Personalized models through transfer learning and domain adaptation techniques
11. Integration into healthcare platforms for mental health monitoring and intervention

12. Conclusion

This project demonstrates an effective Deep CNN-based approach for EEG emotion recognition using the GAMEEMO dataset. The model successfully captures hierarchical spatiotemporal patterns in 14-channel EEG signals and classifies them into four emotion categories with a classification accuracy of 97.06%, surpassing existing approaches including LSTM, Bi-LSTM, and baseline CNN models.

The preprocessing pipeline, comprising band-pass filtering, Z-score normalization, and sliding window segmentation, proved essential for improving signal quality and model performance. The progressive filter depth design (64 to 512 filters) enables the model to extract both low-level signal characteristics and high-level emotional patterns.

The system demonstrates strong potential for real-world applications in mental health monitoring, adaptive learning systems, and human-computer interaction. Future enhancements in real-time processing, wearable integration, and cross-dataset generalization will further strengthen its practical applicability.

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